Power



File: power.kicad_sch

Vehicle (OBD, CAN, K-Line)



File: vehicle.kicad_sch

MCU (ESP32-C5, USB)



File: mcu.kicad_sch

LTE + GNSS (EC200U)



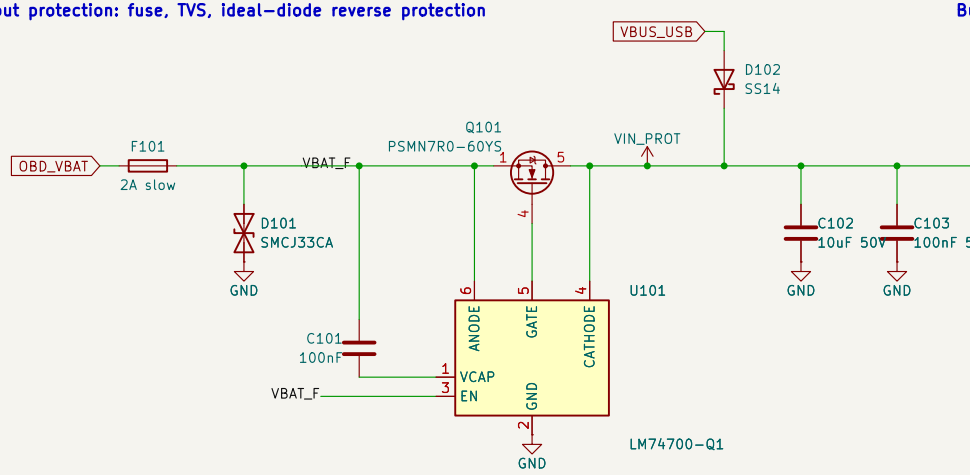
File: lte.kicad_sch

Sheet: /
File: obd-v2x-dongle.kicad_sch

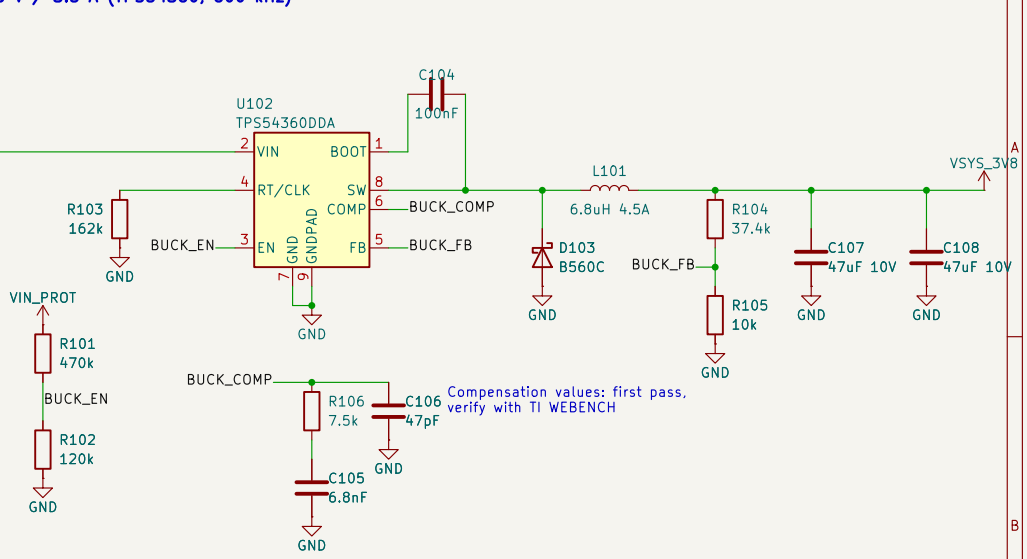
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.8		Id: 1/5

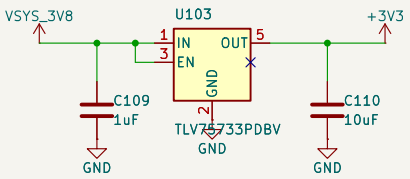
Input protection: fuse, TVS, ideal-diode reverse protection



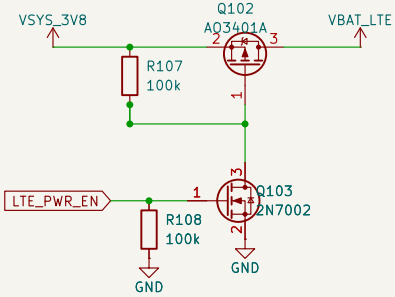
Buck 3.8 V / 3.5 A (TPS54360, 600 kHz)



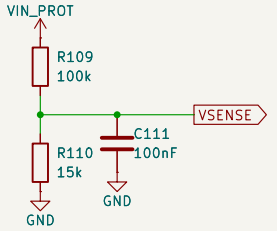
LDO 3.3 V (logic)



LTE modem supply switch (LTE_PWR_EN high = on)



Battery sense (VSENSE = VIN x 0.13)



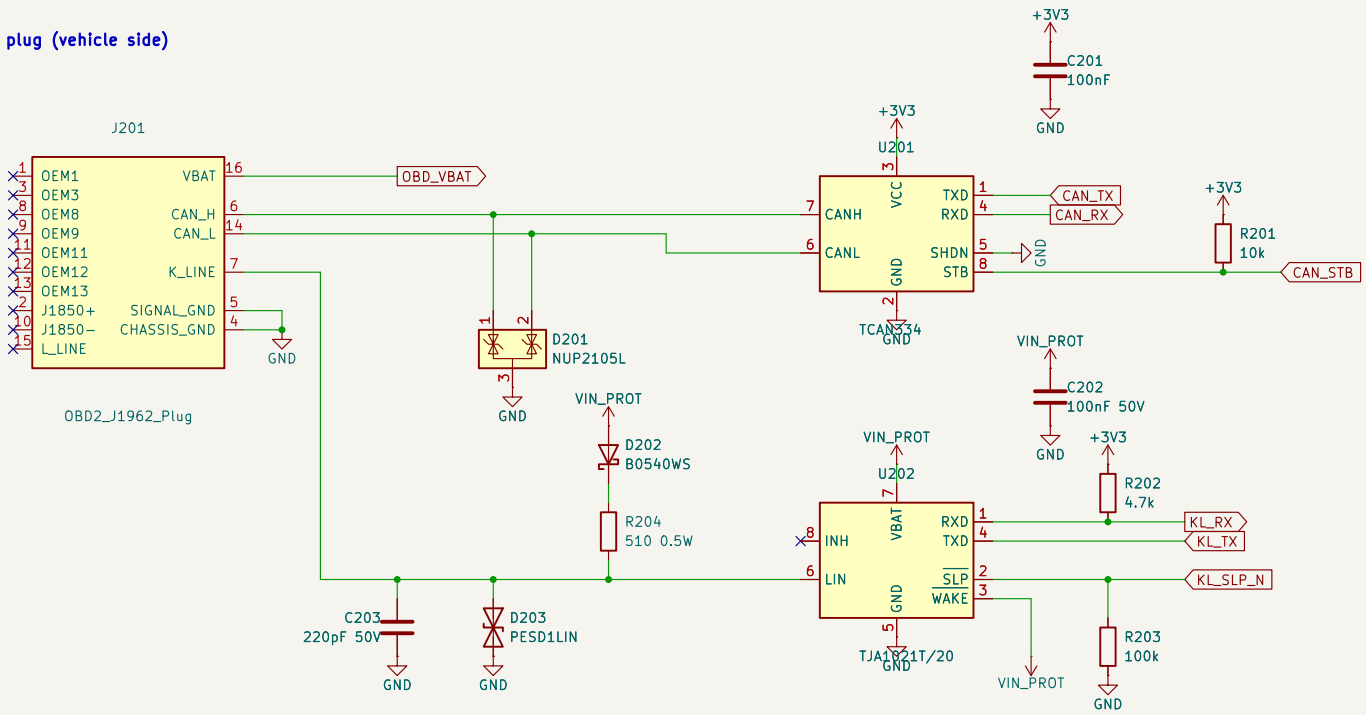
ERC power flags



Sheet: /Power/	
File: power.kicad_sch	
Title: Power: protection, buck 3.8 V, LDO 3.3 V	
Size: A4	Date:
KiCad E.D.A. 10.0.8	**Rev: 0.1**
	Id: 2/5

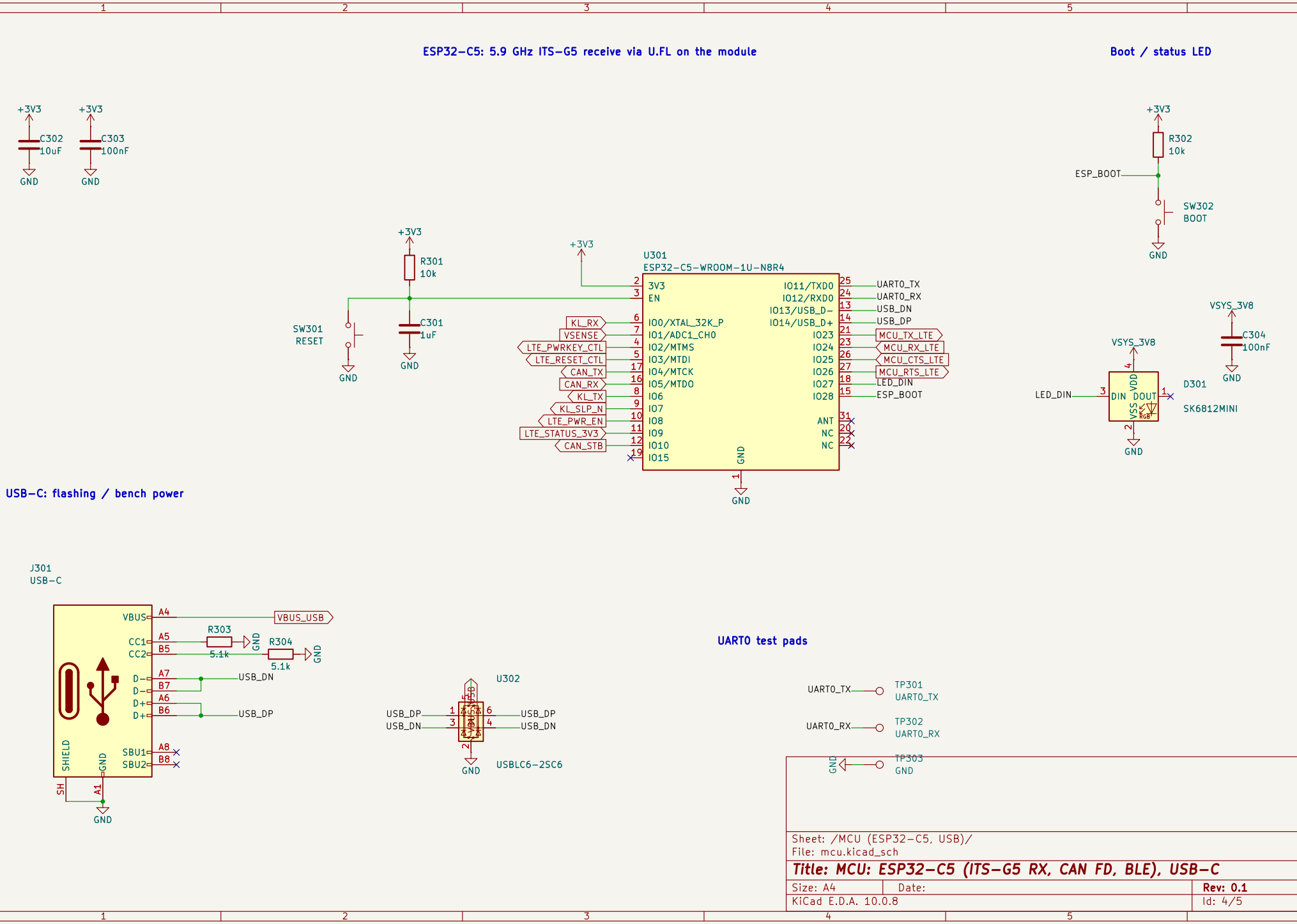
High-speed CAN (ISO 15765), no termination in the dongle

OBD-II plug (vehicle side)

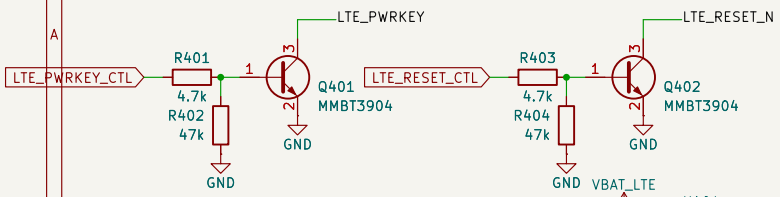


K-Line (ISO 9141 / KWP2000) via LIN transceiver + 510R tester pull-up

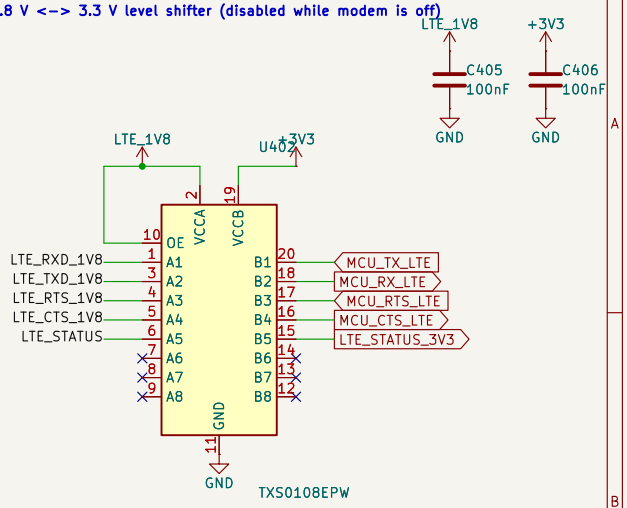
Sheet: /Vehicle (OBD, CAN, K-Line)/	
File: vehicle.kicad_sch	
Title: Vehicle interface: OBD-II, CAN, K-Line	
Size: A4	Date:
KiCad E.D.A. 10.0.8	Rev: 0.1
	Id: 3/5


ESP32-C5: 5.9 GHz ITS-G5 receive via U.FL on the module
Boot / status LED
USB-C: flashing / bench power
UART0 test pads
+3V3
C302
10uF
GND
C303
100nF
R301
10k
SW301
RESET
C301
1uF
U301
ESP32-C5-WROOM-1U-N8R4
3V3
EN
IO0/XTAL_32K_P
IO1/ADC1_CH0
IO2/MTMS
IO3/MTDI
IO4/MTCK
IO5/MTDO
IO6
IO7
IO8
IO9
IO10
IO15
IO11/TXD0
IO12/RXD0
IO13/USB_D-
IO14/USB_D+
IO23
IO24
IO25
IO26
IO27
IO28
ANT
NC
KL_RX
VSENSE
LTE_PWRKEY_CTL
LTE_RESET_CTL
CAN_TX
CAN_RX
KL_TX
KL_SLP_N
LTE_PWR_EN
LTE_STATUS_3V3
CAN_STB
UART0_TX
UART0_RX
USB_DN
USB_DP
MCU_TX_LTE
MCU_RX_LTE
MCU_CTS_LTE
MCU_RTS_LTE
LED_DIN
ESP_BOOT
R302
10k
ESP_BOOT
SW302
BOOT
VSYS_3V8
C304
100nF
D301
SK6812MINI
J301
USB-C
VBUS
CC1
CC2
D-
D+
SBU1
SBU2
SHIELD
SH
R303
5.1k
R304
5.1k
VBUS_USB
U302
USBLC6-2SC6
TP301
UART0_TX
TP302
UART0_RX
TP303
Sheet: /MCU (ESP32-C5, USB)/
File: mcu.kicad_sch
Title: MCU: ESP32-C5 (ITS-G5 RX, CAN FD, BLE), USB-C
Size: A4
Date:
Rev: 0.1
KiCad E.D.A. 10.0.8
Id: 4/5

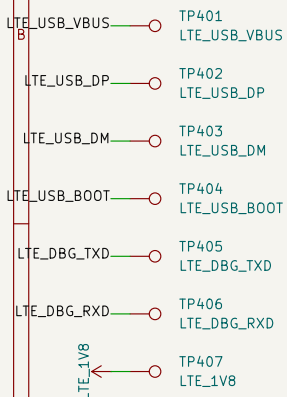
PWRKEY / RESET_N open-collector drivers (MCU high = pull low)

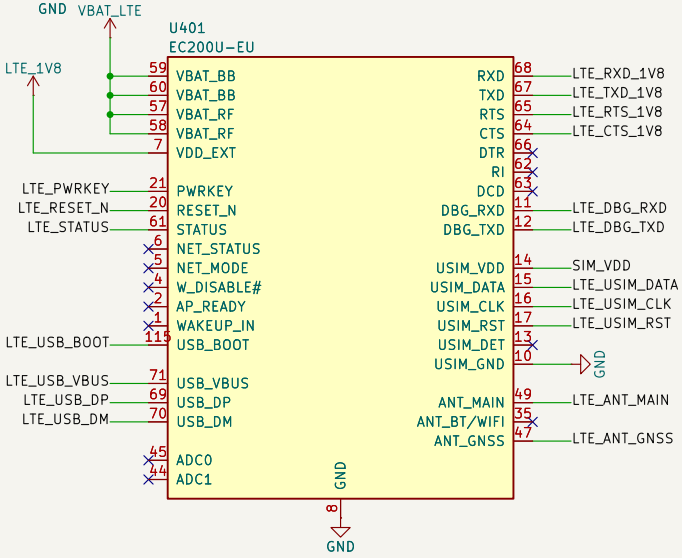


1.8 V <-> 3.3 V level shifter (disabled while modem is off)

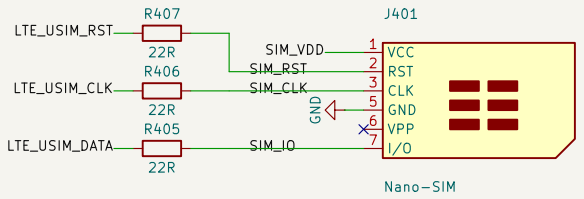


Test pads: modem USB (firmware update), debug UART

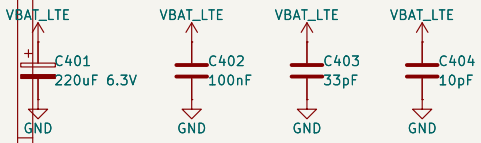


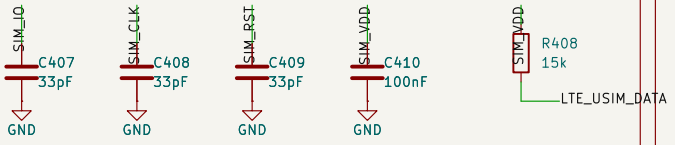


Nano-SIM

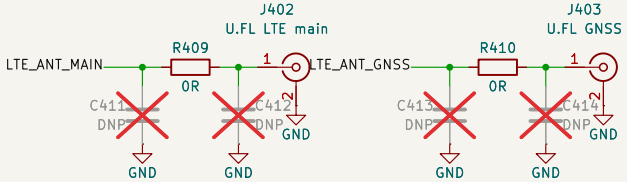


VBAT_LTE bulk/decoupling (close to pins 57-60)





Antennas (pi-match, shunt caps DNP until tuned)



Sheet: /LTE + GNSS (EC200U)/
File: lte.kicad_sch

Title: LTE Cat 1 + GNSS: Quectel EC200U-EU (1.8 V I/O)

Size: A4	Date:	Rev: 0.1
KiCad E.D.A. 10.0.8		Id: 5/5